



HEMANATH T

Aspiring Data Scientist | Machine Learning Enthusiast

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🌐 [Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [LeetCode](#)

PROFILE SUMMARY

Machine Learning Engineer and Data Scientist with hands-on experience in Python, SQL, Machine Learning, Deep Learning, Computer Vision, and Data Analytics. Experienced in developing end-to-end ML pipelines, predictive models, clustering and recommendation systems, and interactive analytics applications. Built and evaluated ML solutions using Scikit-learn, TensorFlow, Pandas, NumPy, OpenCV, YOLO, and RF-DETR, including an object detection system achieving 90.1% mAP@50 and a football analytics platform processing 17K+ player records and 1.8M+ appearance records.

TECHNICAL SKILLS

Programming: Python, SQL

Data Analysis: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing, Feature Engineering, Statistical Analysis

Machine Learning: Regression, Classification, Clustering, PCA, Decision Trees, Random Forest, Ensemble Learning, Model Evaluation

Deep Learning: ANN, CNN, TensorFlow, Transfer Learning

Computer Vision: Object Detection, YOLO, RF-DETR, OpenCV

Visualization: Matplotlib, Seaborn, Plotly, Streamlit

Tools: Git, GitHub, FastAPI, Jupyter Notebook, Google Collab

PROJECTS

AI-Based Rumble Strip Detection System | [GitHub](#)

Tech Stack: Python, Robo-flow, RF-DETR, Computer Vision

- Built and annotated a custom computer vision dataset for highway rumble strip detection using road-scene images.
- Trained and evaluated an RF-DETR object detection model achieving 90.1% mAP@50 on validation data.
- Developed an inference pipeline to detect rumble strips in unseen highway video footage for an early-warning prototype.
- Applied image preprocessing, dataset preparation, object detection, and model evaluation workflows for a real-world road safety application.

Football Player Analytics Dashboard | [Streamlit](#)

Tech Stack: Python, Pandas, Scikit-Learn, Plotly, Streamlit

- Analyzed 17K+ player records and 1.8M+ appearances using Python and Pandas.
- Performed data cleaning, EDA, feature engineering, and player performance analysis.
- Used K-Means clustering and PCA for player segmentation.
- Built a player recommendation engine using performance-based features.
- Developed an interactive Streamlit dashboard with Plotly visualizations.

EDUCATION

M.E COMPUTER SCIENCE AND ENGINEERING - CGPA OF 8.0 / 10

HINDUSTHAN COLLEGE OF ENGINEERING AND TECHNOLOGY | 2024-2026

B.E COMPUTER SCIENCE AND ENGINEERING - CGPA OF 8.6/10

HINDUSTHAN COLLEGE OF ENGINEERING AND TECHNOLOGY | 2019-2023

CERTIFICATIONS & COURSES

Data Science & Analytics Program – ProITbridge

Expected Completion: Sep 2026

- Python, SQL, Statistics, Machine Learning, Deep Learning, Data Visualization (Power BI/Tableau), and Model Deployment.
- Hands-on project: Hazardous Waste Detection using Deep Learning and Object Detection.